

33) Write a system of four equations whose solution gives estimates for the temperatures $T_1, \dots, T_4$.

Soln: ① $4T_1 - T_2 - T_4 = 30$ ② $T_2 = (T_1 + T_3 + 20 + 40)/4$
 or, $4T_2 - T_1 - T_3 = 60$ ③ $T_3 = (T_2 + 40 + T_4 + 30)/4$
 or, $4T_3 - T_2 - T_4 = 70$ ④ $T_4 = (T_1 + T_3 + 10 + 30)/4$, or $4T_4 - T_1 - T_3 = 40$

34) Solve the system of equations from Exercise 33.

Soln: Augmented Matrix:

$$\begin{bmatrix} 4 & -1 & 0 & -1 & 30 \\ 0 & 4 & -1 & 0 & 60 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{bmatrix}$$

Row 1 and 2 are interchanged

Row 2 = Row 2 + 4 × Row 1

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 4 & -1 & 0 & -1 & 30 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{bmatrix}$$

Row 4 = Row 4 - Row 1

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -4 & 0 & 4 & -20 \end{bmatrix}$$

Row 4 = Row 4 × $\frac{1}{4}$

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & 15 & -4 & -1 & 270 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -1 & 0 & 1 & -5 \end{bmatrix}$$

Row 4 and Row 2 are interchanged

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 0 & 1 & -5 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & 15 & -4 & -1 & 270 \end{bmatrix}$$

Row 3 = Row 3 - Row 2

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 0 & 1 & -5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 15 & -4 & -1 & 270 \end{bmatrix}$$

Row 4 = Row 4 + 15 × Row 2

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 0 & 1 & -5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 0 & -4 & 14 & 195 \end{bmatrix}$$

Row 4 = Row 4 + Row 3

$$\begin{bmatrix} -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 0 & 1 & -5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 0 & 0 & 12 & 270 \end{bmatrix}$$